

Lecture 5 — Duality

The price of the resources

Dr. Yahia Djemmada
National Higher School of Mathematics (NHSM), Algiers
yahia.djemmada@nhsm.edu.dz
Operations Research (O.R.)

What you will learn. With the simplex method of Lectures 3 and 4 we can solve every linear program. Today we look again at two numbers of the last tableau of the drone workshop, 25 and 5: how to use them, why they work, and what they are (the value of one more hour). Then we see that they are the solution of a second linear program hidden behind the first one: the *dual problem*. At the end of this lecture you can:

- use the last z-line to prove that a plan is optimal (multiply and add);
- write the dual of any linear program;
- use weak duality to bound the best value, and a dual solution to prove that a plan is optimal;
- read the dual solution (the shadow prices) in the last tableau of the simplex method;
- use the shadow prices to decide: buy more of a resource? make a new product?
- use complementary slackness to test a plan, or to find the dual solution from the primal one.

This lecture starts Chapter 3 of the course (duality and sensitivity analysis).

How this lecture is organised. The lecture has six parts, the same six parts as the slides, with the same titles. Each part starts with a banner that shows this map, so you always know where you are:

1 What are 25 and 5? · 2 The dual problem · 3 Weak and strong duality · 4 Shadow prices, decisions · 5 Complementary slackness · 6 The dual of any problem

We use three examples of the earlier lectures: the drone workshop, the control boards and the spare-part boxes. Definitions, rules, methods and key facts are in boxes, each with the reason why it is true. Each part ends with a short summary (“in short”) and a “Check yourself” question; the answers are at the end.

Contents

Part 1 What are the numbers 25 and 5? 3

1 Where we are: two numbers in the last z-line . . . 3
2 How to use them: multiply and add . . . 4
3 Why it works: the last dictionary is the proof . . . 5
4 What are 25 and 5? The value of the hours . . . 5
5 What do they mean? One more hour . . . 5

Part 2 The dual problem 7

6 Let us consider the following situation . . . 7
7 The best rent: this is the dual problem . . . 7
8 Definition 5.1: the primal and its dual . . . 8
9 Example 5.1: rows become columns . . . 9
10 Why "$\geq$" in the dual constraints? . . . 9
11 The dual of the dual is the primal . . . 10

Part 3 Weak and strong duality 11

12 Key fact 5.1: weak duality . . . 11
13 Consequences of weak duality . . . 11
14 Key fact 5.2: strong duality . . . 12
15 All the cases for a primal–dual pair . . . 12

Part 4 Shadow prices and decisions 14

16 Key fact 5.3: the dual solution in any last tableau . . . 14
17 Shadow prices: checks by solving again . . . 15
18 Using the prices: should we buy more? . . . 16
19 Pricing a new product: drone C . . . 16

Part 5 Complementary slackness 17

20 Key fact 5.4: complementary slackness . . . 17
21 Method 5.1: from the plan to the prices . . . 18
22 Is the corner C $= (4, 18)$ optimal? . . . 18
23 On the picture: the arrow c between the normals . . . 18

Part 6 The dual of any problem 20

24 Rule 5.1: the dual of any problem . . . 20
25 Example 5.2: the dual of the spare-part boxes . . . 21
26 Solving the dual by drawing . . . 21
27 Common mistakes . . . 22

Summary and answers 23

Summary: duality in one picture . . . 23
Answers to "Check yourself" . . . 24

Part 1 of 6

What are the numbers 25 and 5?

1 What are 25 and 5? · 2 The dual problem · 3 Weak and strong duality · 4 Shadow prices, decisions · 5 Complementary slackness · 6 The dual of any problem

In this part, you learn to:

- find the numbers 25 and 5 in the last tableau, and use them: multiply and add;
- see why this proves that the plan is optimal;
- understand what they are: the value of one more hour.

1 Where we are: two numbers in the last z-line

Recall the drone workshop. Each week the workshop has 40 assembly hours, 60 testing hours and 18 cameras. A drone A needs 1 assembly hour and 3 testing hours and brings a profit of 40 thousand DA; a drone B needs 2 assembly hours, 2 testing hours and 1 camera and brings 60 thousand DA:

$$\begin{aligned} \max\ & z = 40x_1 + 60x_2 \\ \text{s.t. } & x_1 + 2x_2 \le 40 && \text{(assembly hours)} \\ & 3x_1 + 2x_2 \le 60 && \text{(testing hours)} \\ & x_2 \le 18 && \text{(cameras)} \\ & x_1, x_2 \ge 0. \end{aligned}$$

The best plan is the corner $\mathrm{B} = (10, 15)$, with $z^* = 1300$ thousand DA (Figure 1). Two numbers appeared twice in the earlier lectures:

- in Lecture 3, the last line of the simplex method was $z = 1300 - 25\,s_1 - 5\,s_2$;
- in Lecture 2, we wrote the line

$$25(x_1 + 2x_2) + 5(3x_1 + 2x_2) = 40x_1 + 60x_2 \le 25 \times 40 + 5 \times 60 = 1300.$$

In this part we answer three questions: *how* do we use these numbers, *why* does it work, and *what* are they?

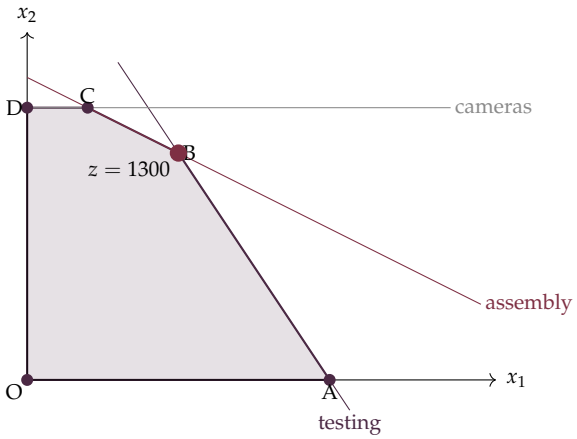


Figure 1: The drone workshop: the best plan is the corner B $= (10, 15)$, where the assembly and testing lines meet.

2 How to use them: multiply and add

After the simplex method we do not choose anything: the last z-line tells us which constraint to multiply, and by which number. Each constraint has its own slack variable (s_1 for assembly, s_2 for testing, s_3 for cameras), and

the number under s_i in the last z-line is the multiplier of constraint i.

A constraint with the number 0 is not used. For the drone workshop, the last z-line is

	x_1	x_2	s_1	s_2	s_3	RHS
z	0	0	**25**	**5**	**0**	**1300**

slack	its constraint	multiply by	result
s_1	assembly: $x_1 + 2x_2 \le 40$	25	$25x_1 + 50x_2 \le 1000$
s_2	testing: $3x_1 + 2x_2 \le 60$	5	$15x_1 + 10x_2 \le 300$
s_3	cameras: $x_2 \le 18$	0	not used
	add		$\mathbf{40x_1 + 60x_2 \le 1300}$

We multiply by *positive* numbers, so each "$\le$" keeps its direction, and two "$\le$" can be added. The left side is exactly the profit, and the right side is the RHS of the z-line. So *every* feasible plan earns at most 1300; the plan B earns 1300, so B is optimal. This is a complete proof, which anybody can check without a drawing and without the tableau.

Another example: the control boards. In TD 2 the last dictionary was $z = 2520 - \frac{9}{2}s_1 - 1\,s_2 - 0\,s_3$. Multiply the machine constraint by $\frac{9}{2}$ and the bench constraint by 1:

$$\tfrac{9}{2}\,(2x_1 + 4x_2) + 1\,(3x_1 + 2x_2) = 12x_1 + 20x_2 \le \tfrac{9}{2} \times 480 + 360 = 2520.$$

3 Why it works: the last dictionary is the proof

Start from the last dictionary of Lecture 3: $z = 1300 - 25\,s_1 - 5\,s_2$. The slacks are the *unused hours*:

$$s_1 = 40 - x_1 - 2x_2, \qquad s_2 = 60 - 3x_1 - 2x_2.$$

Replace them in the dictionary:

$$\begin{aligned} z &= 1300 - 25\,(40 - x_1 - 2x_2) - 5\,(60 - 3x_1 - 2x_2) \\ &= \underbrace{1300 - 1000 - 300}_{=0} + 25\,(x_1 + 2x_2) + 5\,(3x_1 + 2x_2). \end{aligned}$$

So $40x_1 + 60x_2 = 25\,(x_1 + 2x_2) + 5\,(3x_1 + 2x_2)$: this is the line of Lecture 2. And for every feasible plan $s_1 \geq 0$ and $s_2 \geq 0$ (we cannot use more hours than we have), so

$$z = 1300 - \underbrace{25\,s_1}_{\geq 0} - \underbrace{5\,s_2}_{\geq 0} \leq 1300.$$

In short: the long line of Lecture 2 is the last dictionary, written with the *constraints* instead of the slacks. This is true for every optimal tableau: at the optimum the z-line has no negative number, so $z = z^* - (\text{numbers} \geq 0) \times (\text{slacks} \geq 0) \leq z^*$.

4 What are 25 and 5? The value of the hours

Give the value 25 (thousand DA) to one assembly hour and the value 5 to one testing hour. Then:

- a drone A uses 1 assembly hour and 3 testing hours, worth $25 + 3 \times 5 = 40$: exactly its profit;
- a drone B uses 2 assembly hours and 2 testing hours, worth $2 \times 25 + 2 \times 5 = 60$: exactly its profit.

So, for *every* plan,

$$\text{profit} = 25 \times (\text{assembly hours used}) + 5 \times (\text{testing hours used}).$$

Check it on some plans:

plan	assembly hours used	testing hours used	$25 \times$ assembly $+ 5 \times$ testing
A $= (20, 0)$	20	60	$500 + 300 = 800$
$(10, 10)$	30	50	$750 + 250 = 1000$
C $= (4, 18)$	40	48	$1000 + 240 = 1240$
B $= (10, 15)$	**40**	**60**	$1000 + 300 = \mathbf{1300}$

Each value is the real profit of the plan. We have only 40 assembly hours and 60 testing hours, so the profit is at most $25 \times 40 + 5 \times 60 = 1300$, and it reaches 1300 only when *all* the hours are used. This is the meaning of the line of Lecture 2: *you cannot earn more than what your resources are worth*. The cameras get the value 0: at B, 3 cameras are left over.

5 What do they mean? One more hour

Solve the problem again with one hour more or less. While the change is small, the best plan stays where the assembly and testing lines meet, so we solve two equations.

change	new best plan	new z^*	change of z^*
41 assembly hours	$(9.5,\ 15.75)$	1325	$+25$
39 assembly hours	$(10.5,\ 14.25)$	1275	-25
61 testing hours	$(10.5,\ 14.75)$	1305	$+5$
19 cameras	$(10,\ 15)$	1300	0

For the first line: $x_1 + 2x_2 = 41$ and $3x_1 + 2x_2 = 60$; subtracting, $2x_1 = 19$, so $x_1 = 9.5$ and $x_2 = 15.75$; then $z = 40 \times 9.5 + 60 \times 15.75 = 380 + 945 = 1325$. (Fractions of drones are fine here: these are weekly averages.)

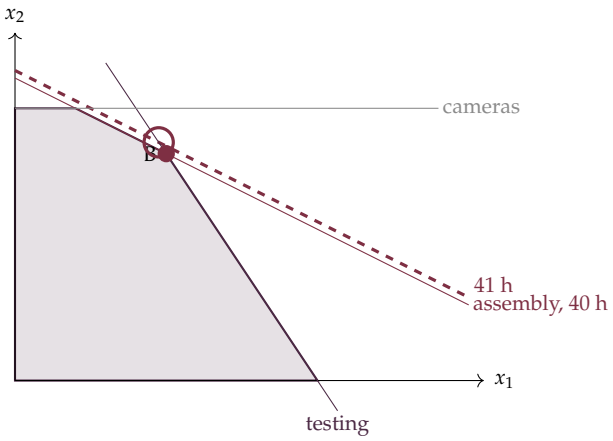


Figure 2: One more assembly hour moves the assembly line a little (dashed). The new best plan $(9.5, 15.75)$ is still where the assembly and testing lines meet, and the profit grows by 25.

So 25 is the *extra profit* given by one more assembly hour, 5 the extra profit of one more testing hour, and 0 for one more camera (3 are already unused).

Cost or income? Neither. The workshop already has its hours, and their cost is not in the model (the profits 40 and 60 are already net). The value 25 says what one hour is *worth* to the workshop. It becomes real money only when hours are traded: buy an extra hour if it costs less than 25, sell one of our hours if someone pays more than 25. Its official name is the *shadow price* (Part 4).

Part 1 in short

After the simplex method, multiply each constraint by the number under its slack in the last z-line and add: the profit is at most the optimal value, so the plan is optimal. This works because the last dictionary, written with the constraints instead of the slacks, is exactly this sum. The numbers are the values of the resources: the profit of every plan is the value of the hours it uses, so no plan can earn more than the value of all the hours. One more assembly hour brings 25 more profit, one more testing hour 5.

Check yourself

Q1. (a) For the drone workshop with 62 testing hours instead of 60, guess the new best profit with the value of a testing hour, then check it by solving $x_1 + 2x_2 = 40$, $3x_1 + 2x_2 = 62$. (b) What happens with 17 cameras instead of 18? Why?

Part 2 of 6

The dual problem

1 What are 25 and 5? — 2 The dual problem — 3 Weak and strong duality — 4 Shadow prices, decisions — 5 Complementary slackness — 6 The dual of any problem

In this part, you learn to:

- look at the workshop from the side of someone who rents its resources;
- write this new problem with variables y_1, y_2, y_3: the dual problem;
- write the dual in canonical form: rows become columns (Definition 5.1).

6 Let us consider the following situation

An investor wants to *rent* all the resources of the workshop for one week: the 40 assembly hours, the 60 testing hours and the 18 cameras. He offers a rent for each unit (in thousand DA):

$y_1 =$ rent of one assembly hour, $\quad y_2 =$ rent of one testing hour, $\quad y_3 =$ rent of one camera,

with $y_1, y_2, y_3 \geq 0$. These are the variables of a new problem.

- The owner accepts the offer only if renting pays *at least* as much as making drones. With the resources of one drone A (1 assembly hour, 3 testing hours) the owner earns 40 by making the drone; so he wants $y_1 + 3y_2 \geq 40$. In the same way, for drone B: $2y_1 + 2y_2 + y_3 \geq 60$.
- The investor pays $40y_1 + 60y_2 + 18y_3$ in total, and he wants to pay as little as possible.

This gives a new linear program, the investor's problem:

$$\begin{aligned} \min\ & w = 40y_1 + 60y_2 + 18y_3 \\ \text{s.t. } & y_1 + 3y_2 \geq 40 && \text{(drone A)} \\ & 2y_1 + 2y_2 + y_3 \geq 60 && \text{(drone B)} \\ & y_1, y_2, y_3 \geq 0. \end{aligned}$$

It has one variable for each resource and one constraint for each drone.

7 The best rent: this is the dual problem

Let us try some rents:

y_1	y_2	y_3	drone A: $y_1 + 3y_2 \geq 40$?	drone B: $2y_1 + 2y_2 + y_3 \geq 60$?	total rent w
0	30	0	90 ✓	60 ✓	1800
40	0	0	40 ✓	80 ✓	1600
30	5	0	45 ✓	70 ✓	1500
40	0	60	40 ✓	140 ✓	2680
20	0	0	20 ×	40 ×	refused
25	5	0	40 ✓	60 ✓	**1300**

The rent $(20, 0, 0)$ is refused: with it, the owner earns more by making drones. The best (smallest) acceptable rent is $y = (25, 5, 0)$, with $w = 1300$. We meet again the numbers 25 and 5 of Part 1, and the *same value* as the best profit (Figure 3): the values of the hours are their fair rents.

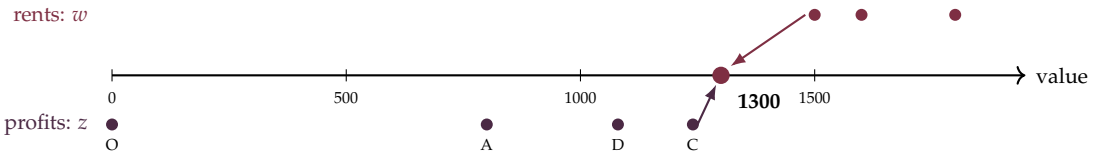


Figure 3: The profits of the corners O, A, D, C (below) and some acceptable rents (above). The profits go up to 1300, the acceptable rents come down to 1300: the two problems meet at 1300.

Names. The investor's problem is called the *dual problem*; the drone problem is the *primal problem*.

8 Definition 5.1: the primal and its dual

Definition 5.1 — Dual problem

The *dual* of the problem (the *primal*)

$$\max\ z = c^\top x \quad \text{s.t.} \quad Ax \le b, \quad x \ge 0$$

is the problem

$$\min\ w = b^\top y \quad \text{s.t.} \quad A^\top y \ge c, \quad y \ge 0.$$

If A has m rows and n columns, the primal has m constraints and n variables, and the dual has n constraints and m variables.

For the drone workshop, side by side:

Primal (the drones)	**Dual** (the rents)
$\max\ 40x_1 + 60x_2$	$\min\ 40y_1 + 60y_2 + 18y_3$
s.t. $x_1 + 2x_2 \le 40$	s.t. $y_1 + 3y_2 \ge 40$
$3x_1 + 2x_2 \le 60$	$2y_1 + 2y_2 + y_3 \ge 60$
$x_2 \le 18, \quad x \ge 0$	$y \ge 0$

How to read the definition:

- the dual has one variable y_i for each primal *constraint* i: the value (rent) of resource i;
- the dual has one constraint for each primal *variable* x_j: the resources used by one unit of product j (column j of A) are worth at least its profit c_j;
- the right-hand side b of the primal becomes the objective of the dual, and the objective c of the primal becomes the right-hand side of the dual;
- the matrix A is *transposed*: its rows become columns; max becomes min, and "$\le$" becomes "$\ge$".

9 Example 5.1: rows become columns

Here is a quick way to write a dual. Write the data in one table: one row for each resource, one column for each product, the right-hand side, and the profits at the bottom. Each row receives its dual variable.

	x_1	x_2	RHS	rent
assembly	1	2	40	y_1
testing	3	2	60	y_2
cameras	0	1	18	y_3
profit	40	60		

The primal reads this table *by rows*: each row is a constraint $\leq$ RHS, and the bottom row is the objective. The dual reads the same table *by columns*:

- column x_1, read downwards: $1y_1 + 3y_2 + 0y_3 \geq 40$;
- column x_2: $2y_1 + 2y_2 + 1y_3 \geq 60$;
- column RHS: min $w = 40y_1 + 60y_2 + 18y_3$;
- and $y_1, y_2, y_3 \geq 0$.

Check the sizes: 2 variables and 3 constraints in the primal; 3 variables and 2 constraints in the dual.

A second example: the control boards. In TD 1 and TD 2 we solved (profit in thousand DA per day)

$$\begin{aligned} \max\ & z = 12x_1 + 20x_2 & & \\ \text{s.t. } & 2x_1 + 4x_2 \leq 480 & & \text{(assembly machine, minutes)} \\ & 3x_1 + 2x_2 \leq 360 & & \text{(test bench, minutes)} \\ & x_2 \leq 100 & & \text{(special sensors),} \qquad x_1, x_2 \geq 0. \end{aligned}$$

With y_1, y_2, y_3 the values of a machine minute, a test bench minute and a sensor, the dual is

$$\begin{aligned} \min\ & w = 480y_1 + 360y_2 + 100y_3 & \\ \text{s.t. } & 2y_1 + 3y_2 \geq 12 & \text{(Basic board)} \\ & 4y_1 + 2y_2 + y_3 \geq 20 & \text{(Pro board)} \\ & y_1, y_2, y_3 \geq 0. & \end{aligned}$$

The multipliers of Section 2, $y = (\frac{9}{2}, 1, 0)$, satisfy the dual constraints ($9 + 3 = 12$ and $18 + 2 = 20$) and give $w = 2160 + 360 = 2520$, the best profit.

10 Why "$\geq$" in the dual constraints?

Come back to the multiply-and-add of Part 1, now with unknown numbers $y_1, y_2, y_3 \geq 0$. Multiplying the three constraints and adding gives, for every feasible plan,

$$(y_1 + 3y_2)\, x_1 + (2y_1 + 2y_2 + y_3)\, x_2 \ \leq\ 40y_1 + 60y_2 + 18y_3.$$

When does this give a bound on the profit $40x_1 + 60x_2$? Two examples:

- $y = (40, 0, 0)$ gives $40x_1 + 80x_2 \le 1600$. In front of x_1 we have $40 \ge 40$, in front of x_2 we have $80 \ge 60$. Since $x_2 \ge 0$, $60x_2 \le 80x_2$, so

$$\text{profit} = 40x_1 + 60x_2 \;\le\; 40x_1 + 80x_2 \;\le\; 1600 :$$

  a true bound.

- $y = (20, 0, 0)$ gives $20x_1 + 40x_2 \le 800$. Now the left side is *smaller* than the profit ($20 < 40$, $40 < 60$), so it tells us nothing about the profit. Indeed B earns $1300 > 800$.

So the dual constraints ask exactly what we need: *in front of each x_j, at least its profit c_j*. Then

$$\text{profit} \;\le\; \text{left side} \;\le\; \text{rent } 40y_1 + 60y_2 + 18y_3.$$

Think of two bags: if my bag weighs at most 10 kg and your bag is lighter than mine, then your bag weighs at most 10 kg. Here the profit is your bag and the left side is my bag.

So every acceptable rent is a bound on the profit, and the dual problem looks for the *best* (smallest) bound. The constraints are "$\ge$" and not "$=$" because a product that is not made may have resources worth *more* than its profit (drone C with profit 28 in Section 19: its resources are worth 30); at the optimum, every product that *is* made gets an equality (Part 5).

11 The dual of the dual is the primal

The dual is a minimisation with "$\ge$" constraints. To take its dual with Definition 5.1, first write it as a maximisation with "$\le$" constraints (multiply the objective and the constraints by -1):

$$\min\; b^\top y,\;\; A^\top y \ge c,\;\; y \ge 0 \quad\Longleftrightarrow\quad \max\,(-b)^\top y,\;\; (-A^\top)\, y \le -c,\;\; y \ge 0.$$

Apply Definition 5.1 to the right-hand problem, calling the new variables x. The matrix $-A^\top$ is transposed again, which gives $-A$:

$$\min\,(-c)^\top x,\;\; (-A)\, x \ge -b,\;\; x \ge 0.$$

Multiply by -1 again: this is max $c^\top x$, $Ax \le b$, $x \ge 0$, the primal. So the primal and the dual form a *pair*: each one is the dual of the other, and we can start from either of them.

Part 2 in short

An investor who rents all the resources offers a rent $y_i \ge 0$ for each unit; the owner accepts only if the resources of each product are paid at least its profit; the investor minimises the total rent. This is the dual problem, and its best rent is $(25, 5, 0)$ with the same value 1300 as the best profit. In general, the dual of max $c^\top x$, $Ax \le b$, $x \ge 0$ is min $b^\top y$, $A^\top y \ge c$, $y \ge 0$ (Definition 5.1): rows become columns. The "$\ge$" of the dual constraints is what makes every acceptable rent a bound on the profit. The dual of the dual is the primal.

Check yourself

Q2. A robot factory makes three robot models. Each robot of model j uses resources R_1, R_2, R_3 as follows, and the profits are in thousand DA:

$$\max\; z = 5x_1 + 4x_2 + 3x_3 \quad \text{s.t.} \quad \begin{aligned} 2x_1 + 3x_2 + x_3 &\le 5 \\ 4x_1 + x_2 + 2x_3 &\le 11 \\ 3x_1 + 4x_2 + 2x_3 &\le 8, \qquad x \ge 0. \end{aligned}$$

Write the dual. How many variables and constraints does it have?

Part 3 of 6

Weak and strong duality

1 What are 25 and 5? — 2 The dual problem — 3 **Weak and strong duality** — 4 Shadow prices, decisions — 5 Complementary slackness — 6 The dual of any problem

In this part, you learn to:

- prove that every plan is below every bound (weak duality, Key fact 5.1);
- use a dual solution as a certificate of optimality;
- know that the best values are equal (strong duality, Key fact 5.2) and all the cases.

12 Key fact 5.1: weak duality

Key fact 5.1 — Weak duality

If x is feasible for the primal ($Ax \le b$, $x \ge 0$) and y is feasible for the dual ($A^\top y \ge c$, $y \ge 0$), then

$$z = c^\top x \;\le\; b^\top y = w.$$

Why is it true? This is the computation of Section 10, written with matrices. We use the four conditions one after the other:

$$c^\top x \;\le\; (A^\top y)^\top x \;=\; y^\top (Ax) \;\le\; y^\top b.$$

- First "$\le$": each $c_j \le (A^\top y)_j$ (dual constraint) and each $x_j \ge 0$, so $c_j x_j \le (A^\top y)_j\, x_j$; add over j.
- The "$=$" in the middle is just $(A^\top y)^\top x = y^\top A x$.
- Second "$\le$": each $(Ax)_i \le b_i$ (primal constraint) and each $y_i \ge 0$, so $y_i (Ax)_i \le y_i b_i$; add over i.

Example. The plan $x = (10, 10)$ is feasible ($30 \le 40$, $50 \le 60$, $10 \le 18$) with $z = 1000$. The prices $y = (30, 5, 0)$ are feasible with $w = 1500$. Indeed $1000 \le 1500$.

13 Consequences of weak duality

Weak duality is a simple inequality, but it has four useful consequences.

1. **A bound for free.** Any dual feasible y gives an upper bound $b^\top y$ on the best profit, without solving anything.
2. **A certificate of optimality.** If x is primal feasible, y is dual feasible and $c^\top x = b^\top y$, then x and y are *both optimal*. Indeed, every feasible plan x' satisfies $c^\top x' \le b^\top y = c^\top x$, so no plan beats x; and every feasible y' satisfies $b^\top y' \ge c^\top x = b^\top y$, so no prices beat y.

   Example: $x = (10, 15)$ and $y = (25, 5, 0)$ are feasible and both give 1300. So B is optimal, and this check takes ten seconds: it is a *certificate*.

3. **Unbounded primal ⇒ infeasible dual.** If the primal is unbounded, z takes values as large as we want. A dual feasible y would bound all of them by $b^\top y$: impossible. So the dual has no feasible solution.
4. **Unbounded dual ⇒ infeasible primal**, for the same reason.

14 Key fact 5.2: strong duality

Key fact 5.2 — Strong duality

If the primal has an optimal solution, then the dual has an optimal solution too, and the optimal values are equal: $z^* = w^*$.

We admit the general proof, but its idea is visible in the last tableau of the simplex method. Here is the last tableau of the drone workshop (Lecture 3):

basis	x_1	x_2	s_1	s_2	s_3	RHS
x_1	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	10
s_3	0	0	$-\frac{3}{4}$	$\frac{1}{4}$	1	3
x_2	0	1	$\frac{3}{4}$	$-\frac{1}{4}$	0	15
z	0	0	25	5	0	1300
			y_1	y_2	y_3	

Figure 4: The last tableau of the drone workshop. Its z-line contains the dual solution $y = (25, 5, 0)$ under the slacks.

Its z-line has no negative number, because the tableau is optimal. Read it with $y = (25, 5, 0)$, the numbers under the slacks:

- under s_1, s_2, s_3: $25, 5, 0 \geq 0$, that is $y \geq 0$;
- under x_1 and x_2: 0 and 0. These numbers are $(A^\top y)_j - c_j$: $25 + 3 \times 5 - 40 = 0$ and $2 \times 25 + 2 \times 5 + 0 - 60 = 0$. They are ≥ 0, that is $A^\top y \geq c$: the dual constraints hold;
- the RHS: $1300 = 40 \times 25 + 60 \times 5 + 18 \times 0 = b^\top y$.

So the last z-line gives a dual feasible y with $w = z^*$. By the certificate of Section 13, this y is optimal for the dual. The same reading works for every optimal tableau of a problem max $c^\top x$, $Ax \leq b$, $x \geq 0$ (Section 16): the simplex method solves the primal and the dual at the same time. This is how strong duality is proved for such problems.

15 All the cases for a primal–dual pair

Each problem of the pair is optimal, unbounded or infeasible (Lecture 2). Weak and strong duality leave only four possible combinations:

	Dual		
Primal	optimal	unbounded	infeasible
optimal	✓ $z^* = w^*$	impossible	impossible
unbounded	impossible	impossible	✓
infeasible	impossible	✓	✓ (possible)

- If one problem is optimal, so is the other, with the same value (strong duality, applied to the primal or to the dual).
- If one problem is unbounded, the other is infeasible (weak duality, Section 13).
- Both problems can be infeasible. Example:

$$\max\ 2x_1 - x_2 \quad \text{s.t.} \quad x_1 - x_2 \le 1,\ \ -x_1 + x_2 \le -2,\ \ x \ge 0.$$

The second constraint says $x_1 - x_2 \ge 2$, and the first says $x_1 - x_2 \le 1$: impossible. Its dual is

$$\min\ y_1 - 2y_2 \quad \text{s.t.} \quad y_1 - y_2 \ge 2,\ \ -y_1 + y_2 \ge -1,\ \ y \ge 0,$$

where the second constraint says $y_1 - y_2 \le 1$: impossible too. This case is rare, but it exists.

A useful guarantee. The dual does not create an optimum: it reflects the situation of the primal. But if you find *one* feasible plan x and *one* feasible y, then neither problem can be unbounded (weak duality) and neither is infeasible, so *both* have an optimal solution, and the best value lies between $c^\top x$ and $b^\top y$. Example: the plan $(10, 10)$ gives 1000 and the rents $(30, 5, 0)$ give 1500, so $1000 \le z^* \le 1500$.

Part 3 in short

Weak duality: every primal feasible value is at most every dual feasible value, $z \le w$ (Key fact 5.1). So a dual feasible y is a bound, and equal values prove that both solutions are optimal (a certificate). Strong duality: when the primal has an optimum, so does the dual, with the same value (Key fact 5.2); the simplex method finds both. Unbounded on one side means infeasible on the other; both can be infeasible.

Check yourself

Q3. In the drone workshop, check that the plan $(12, 12)$ is feasible and that the rents $y = (20, 10, 0)$ are acceptable (dual feasible). Without solving anything, what can you say about the best profit z^*? A colleague claims a plan that earns 1450: is it possible?

Part 4 of 6

Shadow prices and decisions

1 What are 25 and 5? — 2 The dual problem — 3 Weak and strong duality — 4 **Shadow prices, decisions** — 5 Complementary slackness — 6 The dual of any problem

In this part, you learn to:

- read the dual solution in any last tableau, with slacks and surpluses (Key fact 5.3);
- check the shadow prices by solving again (Definition 5.2);
- use them to decide: buy or sell a resource? make a new product?

16 Key fact 5.3: the dual solution in any last tableau

In Part 1 we read the numbers 25 and 5 under the slacks of a maximisation with "$\leq$" constraints. The same reading works with surplus variables and for minimisations; only the sign needs care. Recall the words of Rule 5.1 (Part 6): a constraint is *natural* if it is "$\leq$" in a max problem or "$\geq$" in a min problem (a resource, or a need), and *strange* otherwise.

Key fact 5.3 — Reading the dual solution

Solve the problem with the simplex method (a minimisation is solved as the maximisation of $-z$). In the last z-line, let $d_i \geq 0$ be the number under the slack or the surplus of constraint i. Then the optimal dual variable is

$$y_i = d_i \text{ if constraint } i \text{ is natural}, \qquad y_i = -d_i \text{ if constraint } i \text{ is strange}.$$

For max $c^\top x$, $Ax \leq b$, $x \geq 0$ (all natural), y_i is simply the number under s_i, and the number under x_j is the *dual surplus* $(A^\top y)_j - c_j \geq 0$: the value of the resources of product j minus its profit.

Why, for max $c^\top x$, $Ax \leq b$, $x \geq 0$*?* The z-line of the first tableau is $z - c^\top x = 0$, with 0 under the slacks. Each pivot adds to the z-line a multiple of a constraint line. After all the pivots, the z-line is the first z-line plus some combination $y_1 L_1 + \cdots + y_m L_m$ of the *original* constraint lines L_i (each pivot line is itself a combination of the original lines). In the original lines, the column of s_i has a 1 in line i and 0 elsewhere, so the number under s_i at the end is exactly y_i. Under x_j we get $-c_j + \sum_i y_i a_{ij} = (A^\top y)_j - c_j$, and on the right $\sum_i y_i b_i = b^\top y$. At the optimum all these numbers are ≥ 0: this is the reading of Section 14. The other cases follow from the same argument, with two possible changes of sign: a surplus has the coefficient -1 in its line, and a minimisation is solved with $-z$. A natural constraint has no change or two changes, which cancel (a "$\geq$" in a min); a strange constraint has exactly one, hence $y_i = -d_i$.

Examples.

problem	last z-line	dual solution
drone workshop (max, all $\leq$)	25, 5, 0 under s_1, s_2, s_3	$y = (25, 5, 0)$
control boards (max, all $\leq$)	$\frac{9}{2}$, 1, 0 under s_1, s_2, s_3	$y = (\frac{9}{2}, 1, 0)$
spare-part boxes (min, all $\geq$)	$\frac{20}{3}$, $\frac{50}{3}$ under e_1, e_2	$y = (\frac{20}{3}, \frac{50}{3})$
drones + order $x_2 \geq 16$ (max)	40 under s_1, 20 under e_4	$y_1 = 40$, $y_4 = -20$

(The control boards are in TD 2; the spare-part boxes and the order are in Lecture 4, the order in its Check yourself Q4.) In the last example, the drone workshop must make at least 16 drones B; the optimum is $(8, 16)$ with $z^* = 1280$, and the last dictionary is $z = 1280 - 40\,s_1 - 20\,e_4$. The order is a "$\geq$" in a max problem: strange, so $y_4 = -20$.

Using the numbers for a proof. Turn every constraint in the direction of the bound ("$\leq$" for a max, "$\geq$" for a min) by multiplying the strange ones by -1; then multiply each constraint by its $d_i \geq 0$ and add. For the order example: $40 \times (x_1 + 2x_2 \leq 40)$ gives $40x_1 + 80x_2 \leq 1600$; the order $x_2 \geq 16$ becomes $-x_2 \leq -16$, and $20\times$ it gives $-20x_2 \leq -320$; adding, $40x_1 + 60x_2 \leq 1280$. For the spare-part boxes, both constraints are already "$\geq$":

$$\tfrac{20}{3}\,(2x_1 + x_2) + \tfrac{50}{3}\,(x_1 + 2x_2) = 30x_1 + 40x_2 \;\geq\; \tfrac{20}{3} \times 8 + \tfrac{50}{3} \times 7 = 170,$$

a *lower* bound on the cost. As in Section 3, this is the last dictionary written with the constraints: $z = z^* - \sum(\text{numbers} \geq 0) \times (\text{slacks and surpluses} \geq 0)$.

An "=" constraint has no slack and no surplus (its artificial column was deleted after Phase 1). Its dual variable, which can have any sign, is found with Method 5.1 (Part 5).

17 Shadow prices: checks by solving again

Definition 5.2 — Shadow price

The *shadow price* of constraint i is the change of the best value when its right-hand side b_i increases by one unit (for small changes). It is equal to the optimal dual variable y_i.

Let us check it by solving the problems again.

problem	change	new best plan	new best value	change
drone workshop	41 assembly hours	(9.5, 15.75)	profit 1325	$+25 = y_1$ ✓
drone workshop	19 cameras	(10, 15)	profit 1300	$0 = y_3$ ✓
spare-part boxes	9 motors needed	$(\frac{11}{3}, \frac{5}{3})$	cost $\frac{530}{3} \approx 176.67$	$+\frac{20}{3} = y_1$ ✓
drones + order	order of 17 drones B	(6, 17)	profit 1260	$-20 = y_4$ ✓

For the spare-part boxes: $2x_1 + x_2 = 9$ and $x_1 + 2x_2 = 7$ give $x_1 = \frac{11}{3}$, $x_2 = \frac{5}{3}$, and the cost $30 \times \frac{11}{3} + 40 \times \frac{5}{3} = 110 + \frac{200}{3} = \frac{530}{3} = 170 + \frac{20}{3}$. For the order: $x_2 = 17$ and $x_1 = 40 - 34 = 6$ give $240 + 1020 = 1260$.

The signs make sense. A natural constraint is a resource or a need: one more unit of a resource increases the profit, one more unit of a need increases the cost, and $y_i \geq 0$. A strange constraint is a limit that hurts: ordering one more drone B *costs* 20 of profit, so $y_4 = -20$.

How far? A shadow price is valid only while the best plan stays at the meeting point of the same lines. With many more assembly hours, for example, the meeting point of the assembly and testing

lines leaves the feasible set (it crosses the camera line), and the price changes. Finding how far we can go is *sensitivity analysis*: the subject of Lecture 6.

18 Using the prices: should we buy more?

A partner offers extra resources. We compare his price with the shadow price:

- one more assembly hour for 20 thousand DA? It brings 25: *yes*, we gain $25 - 20 = 5$;
- one more testing hour for 8? It brings 5: *no*, we would lose $8 - 5 = 3$;
- one more camera for 1? It brings 0: *no*, we already have 3 unused cameras;
- the other way: the partner wants to *buy* one of our testing hours. We lose 5 if we sell it, so we accept only a price above 5.

Rule of thumb. A resource is worth its shadow price: buy below it, sell above it (for small quantities). A resource that is not fully used is worth 0.

19 Pricing a new product: drone C

The workshop studies a new drone C. It needs 1 assembly hour, 1 testing hour and 1 camera, and brings a profit p. Is it worth making?

At the shadow prices, the resources of one drone C are worth

$$1 \times 25 + 1 \times 5 + 1 \times 0 = 30 \text{ thousand DA.}$$

- If $p > 30$, drone C earns more than the value of the resources it takes from A and B: it is worth making. For example, with $p = 35$ the new best plan is $(x_1, x_2, x_3) = (10, 12, 6)$ with $z^* = 1330 > 1300$. (Check: assembly $10 + 24 + 6 = 40$, testing $30 + 24 + 6 = 60$, cameras $12 + 6 = 18$; profit $400 + 720 + 210 = 1330$.)
- If $p < 30$, it is not worth making: with $p = 28$ the best plan stays B with $z^* = 1300$.
- If $p = 30$, making some drones C neither increases nor decreases the best profit.

In the language of duality, the new product adds a new dual constraint $y_1 + y_2 + y_3 \geq p$. If the current prices satisfy it, they stay optimal and nothing changes; if not ($30 < p$), the product must enter the plan. This test is called *pricing*. Note that after adding C with $p = 35$, the cameras are all used and get a positive price: the new dual solution is $y = (17.5, 7.5, 10)$, with $w = 700 + 450 + 180 = 1330$.

Part 4 in short

In any optimal tableau, the dual solution is read under the slacks and surpluses: $y_i = d_i$ for a natural constraint, $y_i = -d_i$ for a strange one (Key fact 5.3). Each y_i is a shadow price: the change of the best value for one more unit of b_i, for small changes (Definition 5.2). Buy a resource below its shadow price, sell above it; a resource not fully used is worth 0. A new product is worth making only if its profit is more than the value of its resources at the shadow prices (pricing).

Check yourself

Q4. Control boards: the shadow prices are $y = (\frac{9}{2}, 1, 0)$. A partner offers 10 extra machine minutes per day for 30 thousand DA per day in total. Is it a good deal? Check your answer

by solving the two equations of the machine and the test bench with $b_1 = 490$.

Part 5 of 6

Complementary slackness

1 What are 25 and 5? · 2 The dual problem · 3 Weak and strong duality · 4 Shadow prices, decisions · 5 **Complementary slackness** · 6 The dual of any problem

In this part, you learn to:

- state complementary slackness (Key fact 5.4);
- find the dual solution from a primal solution, or show that a plan is not optimal (Method 5.1);
- see it on the picture.

20 Key fact 5.4: complementary slackness

Key fact 5.4 — Complementary slackness

Let x be primal feasible and y dual feasible. Then x and y are both optimal if and only if

1. for each primal constraint i: $y_i = 0$ or constraint i is binding ($s_i = 0$);
2. for each primal variable j: $x_j = 0$ or dual constraint j is an equality ($(A^\top y)_j = c_j$).

Why is it true? Look again at the chain of weak duality:

$$c^\top x \;\le\; y^\top A x \;\le\; y^\top b.$$

By strong duality and the certificate, x and y are both optimal exactly when $c^\top x = b^\top y$, that is when *both* "$\le$" are "$=$".

- The second gap is $y^\top b - y^\top A x = \sum_i y_i\,(b_i - (Ax)_i) = \sum_i y_i\, s_i$. Each term is a product of two non-negative numbers. The sum is 0 only if each product $y_i s_i$ is 0: condition 1.
- The first gap is $y^\top A x - c^\top x = \sum_j x_j\,((A^\top y)_j - c_j)$. Again each term is ≥ 0, so the sum is 0 only if each product is 0: condition 2.

In words. A resource that is *not fully used* has price 0. A product that is *made* earns exactly the value of its resources at the optimal prices: no more (otherwise the prices would not be feasible) and no less (otherwise it should not be made).

Drone workshop. At B, $s_3 = 3 > 0$ and indeed $y_3 = 0$. Both drones are made ($x_1, x_2 > 0$), and both dual constraints are equalities: $25 + 15 = 40$ and $50 + 10 + 0 = 60$.

21 Method 5.1: from the plan to the prices

Method 5.1 — Test a plan with complementary slackness

Let x be a feasible plan.

1. Compute the slacks. For each constraint with $s_i > 0$, set $y_i = 0$.
2. For each variable with $x_j > 0$, write dual constraint j as an *equation*.
3. Solve these equations for the remaining y_i.
4. If the solution y is dual feasible ($y \geq 0$ and all the dual constraints hold), then x and y are both optimal. If the equations have only one solution and it is not dual feasible, then x is *not* optimal.

Test of B $= (10, 15)$.

1. Slacks: assembly $10 + 30 = 40$, so $s_1 = 0$; testing $30 + 30 = 60$, so $s_2 = 0$; cameras $s_3 = 18 - 15 = 3 > 0$, so $y_3 = 0$.
2. $x_1 = 10 > 0$: $y_1 + 3y_2 = 40$. $x_2 = 15 > 0$: $2y_1 + 2y_2 + y_3 = 60$, that is $y_1 + y_2 = 30$ (with $y_3 = 0$).
3. Subtract: $2y_2 = 10$, so $y_2 = 5$ and $y_1 = 25$.
4. $y = (25, 5, 0) \geq 0$ and both dual constraints hold (as equalities). So B is optimal, and $w = 1000 + 300 = 1300 = z$.

We found the dual solution without any tableau, only from the plan.

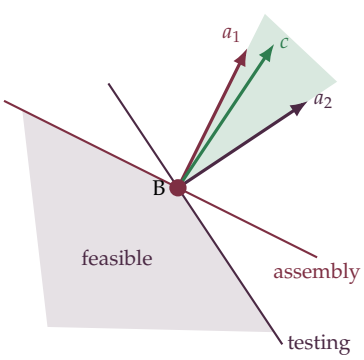
22 Is the corner C $= (4, 18)$ optimal?

1. Slacks at C: assembly $4 + 36 = 40$, so $s_1 = 0$; testing $12 + 36 = 48$, so $s_2 = 12 > 0$ and $y_2 = 0$; cameras $18 = 18$, so $s_3 = 0$.
2. $x_1 = 4 > 0$: $y_1 + 3y_2 = 40$, so $y_1 = 40$.
3. $x_2 = 18 > 0$: $2y_1 + 2y_2 + y_3 = 60$, so $80 + 0 + y_3 = 60$ and $y_3 = -20$.
4. The equations have only one solution, $y = (40, 0, -20)$, and it is not dual feasible ($y_3 < 0$). So C is *not* optimal. Indeed, C gives $z = 160 + 1080 = 1240 < 1300$.

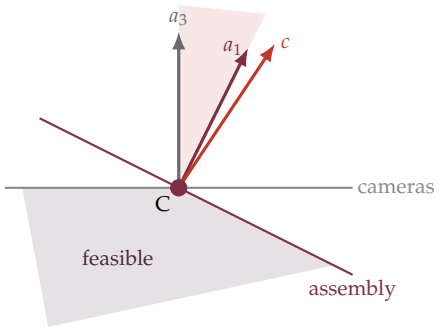
The negative price has a meaning: at C, the cameras are a burden, not a treasure. Using one camera less (moving from C along the assembly line towards B) increases the profit.

23 On the picture: the arrow c between the normals

The rows of A, $a_1 = (1, 2)$, $a_2 = (3, 2)$, $a_3 = (0, 1)$, are perpendicular (normal) to the assembly, testing and camera lines, and they point out of the feasible set. The dual constraints say $y_1 a_1 + y_2 a_2 + y_3 a_3 \geq c$, and at the optimum, by complementary slackness, they are equalities for the products made, with $y_i = 0$ for the lines that are not binding. So at an optimal corner, c is a combination of the normals of the binding lines, with prices ≥ 0.



At B: $c = 25\,a_1 + 5\,a_2$ ✓



At C: $c = 40\,a_1 - 20\,a_3$ ×

Figure 5: Zoom on the corners B and C. The arrows a_i are the normals of the binding lines; $c = (40, 60)$ is the objective arrow (drawn shorter). At B, c lies between a_1 and a_2 (green region): B is optimal. At C, c lies outside the region between a_1 and a_3 (red): C is not optimal.

- At B, the binding lines are assembly and testing, and $c = (40, 60) = 25\,(1, 2) + 5\,(3, 2)$: positive prices, c lies between the two normals. Every direction d that stays feasible satisfies $a_1^\top d \le 0$ and $a_2^\top d \le 0$, so $c^\top d = 25\,a_1^\top d + 5\,a_2^\top d \le 0$: no feasible move increases the profit.
- At C, the binding lines are assembly and cameras, and $c = 40\,(1, 2) - 20\,(0, 1)$: a negative price, c lies outside. The direction $d = (2, -1)$ (along the assembly line towards B) stays feasible and $c^\top d = 80 - 60 = 20 > 0$: the profit increases.

Part 5 in short

Complementary slackness (Key fact 5.4): feasible x and y are both optimal exactly when every resource not fully used has price 0 and every product made earns exactly the value of its resources. Method 5.1 uses it to find the dual solution from a plan, or to prove that a plan is not optimal. On the picture: a corner is optimal when c is a combination of the normals of its binding lines with prices ≥ 0.

Check yourself

Q5. For the robot factory of Q2, use Method 5.1 to check that the plan $x = (2, 0, 1)$ is optimal, and give the shadow prices of the three resources.

Part 6 of 6

The dual of any problem

1 What are 25 and 5? — 2 The dual problem — 3 Weak and strong duality — 4 Shadow prices, decisions — 5 Complementary slackness — 6 The dual of any problem

In this part, you learn to:

- write the dual of a minimisation, of "$\geq$" and "$=$" constraints, of free variables (Rule 5.1);
- solve a small dual by drawing and compare with the primal;
- read the meaning of the dual of a cost problem.

24 Rule 5.1: the dual of any problem

Until now the primal was a maximisation in canonical form. Rule 5.1 gives the dual of any linear program. What never changes: one dual variable for each constraint, one dual constraint for each variable, b and c exchange their roles, and A is transposed.

Rule 5.1 — Writing the dual

Max problem		Min problem
constraint i is "$\leq$"	$\longleftrightarrow$	$y_i \geq 0$
constraint i is "$\geq$"	$\longleftrightarrow$	$y_i \leq 0$
constraint i is "$=$"	$\longleftrightarrow$	y_i free (any sign)
$x_j \geq 0$	$\longleftrightarrow$	constraint j is "$\geq$"
$x_j \leq 0$	$\longleftrightarrow$	constraint j is "$\leq$"
x_j free	$\longleftrightarrow$	constraint j is "$=$"

If the primal is a maximisation, read the table from left to right; if it is a minimisation, read it from right to left.

How to remember it. "Natural" goes with "natural": in a max problem a "$\leq$" constraint is natural (a resource) and gives a natural price $y_i \geq 0$; this is Definition 5.1. "Strange" goes with "strange": a "$\geq$" constraint in a max problem gives $y_i \leq 0$. An equality gives a free variable.

Why? Bring the problem to canonical form, apply Definition 5.1, and come back:

- A "$\geq$" constraint $a^\top x \geq \beta$ in a max problem is $-a^\top x \leq -\beta$, with a price $y' \geq 0$. In the dual, this price always appears with the coefficients $-a$ and $-\beta$; writing $y_i = -y'$ we get the coefficients a and β and $y_i \leq 0$.
- An equality $a^\top x = \beta$ is two constraints, $a^\top x \leq \beta$ (price $y' \geq 0$) and $-a^\top x \leq -\beta$ (price $y'' \geq 0$). They always appear together as $y_i = y' - y''$, which can have any sign.
- The rules for the variables follow in the same way, or from the fact that the dual of the dual is the primal.

Example. max $2x_1 + 3x_2$ s.t. $x_1 + x_2 = 4$, $x_1 - 2x_2 \geq -2$, $x_1 \geq 0$, x_2 free. Read from left to right: the equality gives y_1 free, the "$\geq$" gives $y_2 \leq 0$; $x_1 \geq 0$ gives a "$\geq$" constraint (column of x_1: coefficients 1 and 1), x_2 free gives an equality (column of x_2: coefficients 1 and -2):

$$\min\ w = 4y_1 - 2y_2 \quad \text{s.t.} \quad y_1 + y_2 \geq 2,\ \ y_1 - 2y_2 = 3,\ \ y_1 \text{ free},\ y_2 \leq 0.$$

(The primal optimum is $(2, 2)$ with $z^* = 10$; the dual optimum is $y = (\frac{7}{3}, -\frac{1}{3})$ with $w^* = \frac{28}{3} + \frac{2}{3} = 10$. The "strange" constraint really gets a negative price.)

25 Example 5.2: the dual of the spare-part boxes

Recall the spare-part boxes (Lectures 2 and 4). A box A contains 2 motors and 1 sensor and costs 30 thousand DA; a box B contains 1 motor and 2 sensors and costs 40 thousand DA; we need at least 8 motors and 7 sensors:

$$\begin{aligned} \min\ & z = 30x_1 + 40x_2 \\ \text{s.t. } & 2x_1 + x_2 \geq 8 \qquad \text{(motors)} \\ & x_1 + 2x_2 \geq 7 \qquad \text{(sensors)}, \qquad x_1, x_2 \geq 0. \end{aligned}$$

This is a minimisation, so we read Rule 5.1 from right to left. Its constraints are "$\geq$": natural for a minimisation, so $y_1, y_2 \geq 0$. Its variables are ≥ 0, so the dual constraints are "$\leq$". With y_1 and y_2 the prices of one motor and one sensor:

$$\begin{aligned} \max\ & w = 8y_1 + 7y_2 \\ \text{s.t. } & 2y_1 + y_2 \leq 30 \quad \text{(box A)} \\ & y_1 + 2y_2 \leq 40 \quad \text{(box B)}, \qquad y_1, y_2 \geq 0. \end{aligned}$$

The meaning. A supplier sells motors and sensors separately, at prices y_1 and y_2. His prices must not make a box more expensive than buying its parts (otherwise we would keep buying boxes): $2y_1 + y_2 \leq 30$ and $y_1 + 2y_2 \leq 40$. He wants the largest income for the 8 motors and 7 sensors we need: max $8y_1 + 7y_2$. Weak duality now says: every price list of the supplier gives at most the cost of every plan of boxes.

26 Solving the dual by drawing

The dual has two variables, so we can solve it with the method of Lecture 2 (Figure 6). The feasible set has four corners:

corner	how	$w = 8y_1 + 7y_2$
$(0, 0)$	origin	0
$(15, 0)$	$2y_1 + y_2 = 30,\ y_2 = 0$	120
$(\frac{20}{3}, \frac{50}{3})$	$2y_1 + y_2 = 30,\ y_1 + 2y_2 = 40$	$\frac{160}{3} + \frac{350}{3} = \mathbf{170}$
$(0, 20)$	$y_1 + 2y_2 = 40,\ y_1 = 0$	140

For the third corner: from $2y_1 + y_2 = 30$, $y_2 = 30 - 2y_1$; then $y_1 + 60 - 4y_1 = 40$, so $y_1 = \frac{20}{3}$ and $y_2 = 30 - \frac{40}{3} = \frac{50}{3}$.

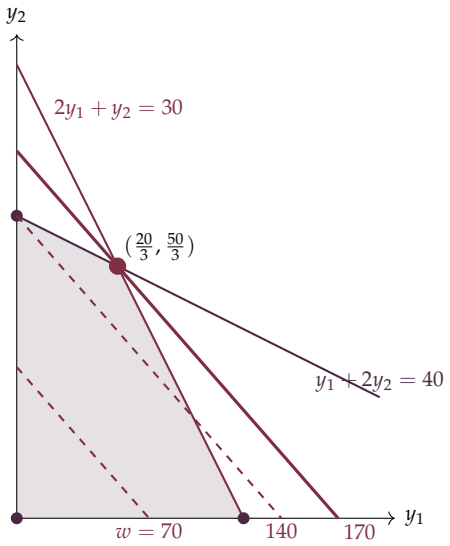


Figure 6: The dual of the spare-part boxes. The level lines $8y_1 + 7y_2 = w$ move up; the last point of the feasible set that they touch is $(\frac{20}{3}, \frac{50}{3})$, with $w^* = 170$.

So the best prices are about 6.67 thousand DA for a motor and 16.67 for a sensor, and $w^* = 170$. The primal optimum is $(3, 2)$: 3 boxes A and 2 boxes B, cost $90 + 80 = 170$. Equal values, as strong duality says. And these prices are the numbers $\frac{20}{3}$ and $\frac{50}{3}$ under the surplus variables in the last tableau of Lecture 4.

Complementary slackness. Both kinds of boxes are bought ($x_1, x_2 > 0$), so both dual constraints are equalities: this is exactly the corner $(\frac{20}{3}, \frac{50}{3})$. Both needs are binding ($6 + 2 = 8$ motors, $3 + 4 = 7$ sensors), which allows both prices to be positive.

Meaning. If we needed one more motor (9 instead of 8), the best cost would increase by about 6.67 thousand DA; one more sensor, by about 16.67. These are the shadow prices of the needs.

27 Common mistakes

Wrong ×	Right ✓
Read the matrix A by rows to write the dual constraints	One dual constraint per *column* of A (per primal variable)
Keep c in the dual objective	The dual objective uses b, the right-hand sides of the primal
Give the sign $y_i \geq 0$ to every dual variable	Use Rule 5.1: "$\geq$" in a max problem gives $y_i \leq 0$, "$=$" gives y_i free
Read the dual solution under x_1, x_2 in the last z-line	The dual solution is under the slacks and surpluses (Key fact 5.3, with its sign rule)
Believe that a shadow price holds for any change of b_i	Only for small changes; how far: sensitivity analysis (Lecture 6)
Give a positive price to a resource that is not fully used	Complementary slackness: $s_i > 0$ gives $y_i = 0$
Think that $z \leq w$ for two *optimal* values only	Weak duality holds for *every* feasible x and y

Part 6 in short

Rule 5.1 gives the dual of any problem: natural constraints ("$\leq$" in a max, "$\geq$" in a min) give natural variables (≥ 0), strange ones give ≤ 0, equalities give free variables, and the same for the variables. The dual of a cost problem (Example 5.2) gives a price to each need. A small dual can be solved by drawing; the primal and dual optimal values are equal.

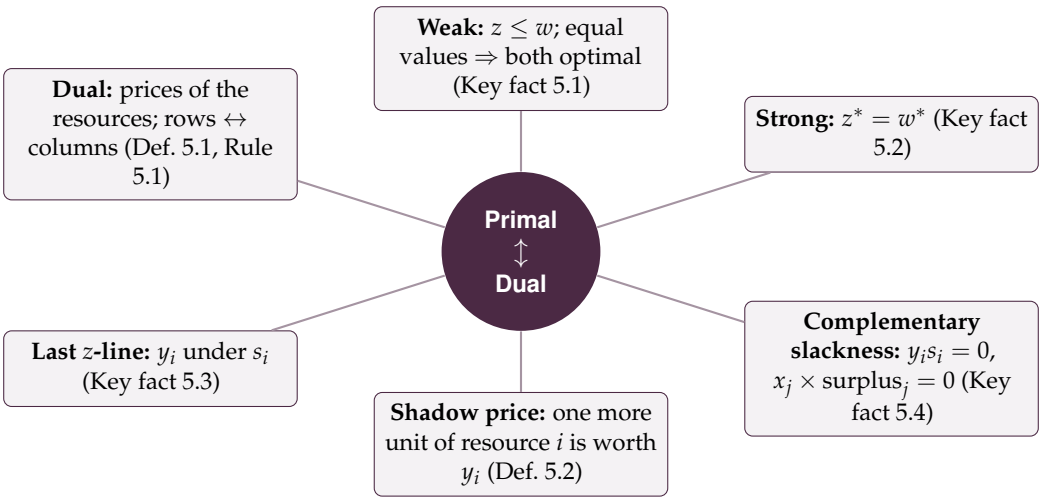
Check yourself

Q6. Write the dual of

$$\min\ z = 4x_1 + 3x_2 \quad \text{s.t.} \quad x_1 + x_2 \geq 3,\ \ x_1 - x_2 \leq 1,\ \ x_1, x_2 \geq 0.$$

The primal optimum is $(0, 3)$ with $z^* = 9$. Use complementary slackness to find the dual solution, and check that $w^* = 9$.

Summary: duality in one picture



- The dual of max $c^\top x$, $Ax \leq b$, $x \geq 0$ is min $b^\top y$, $A^\top y \geq c$, $y \geq 0$ (Definition 5.1); for any problem, use Rule 5.1. The dual of the dual is the primal.
- Weak duality: $c^\top x \leq b^\top y$ for all feasible x and y (Key fact 5.1). Equal values prove that both are optimal.
- Strong duality: if one problem has an optimum, both have one, with the same value (Key fact 5.2).
- In the optimal tableau, y_i is the number of the z-line under s_i (Key fact 5.3); y_i is the shadow price of resource i (Definition 5.2).
- Complementary slackness: $y_i s_i = 0$ and $x_j \times (\text{dual surplus})_j = 0$ (Key fact 5.4); Method 5.1 tests a plan.
- Use the prices to decide: buy a resource below its shadow price; make a new product only if its profit is more than the value of its resources.

What comes next. A shadow price is valid only for small changes. How far can a resource b_i or a profit c_j change before the best plan changes? This is *sensitivity analysis*, the subject of Lecture 6.

Reading. F. S. Hillier and G. J. Lieberman, *Introduction to Operations Research*, Chapter "Duality theory", Sections 6.1 to 6.4; H. A. Taha, *Operations Research: An Introduction*, Chapter "Duality and post-optimal analysis", Sections 4.1 to 4.3.

Answers to "Check yourself"

Q1 (Part 1). (a) A testing hour is worth 5, so 2 more testing hours should bring $2 \times 5 = 10$: guess $z^* = 1310$. Check: subtract $x_1 + 2x_2 = 40$ from $3x_1 + 2x_2 = 62$: $2x_1 = 22$, so $x_1 = 11$ and $x_2 = (40 - 11)/2 = 14.5$; the cameras: $14.5 \leq 18$ ✓. Profit $40 \times 11 + 60 \times 14.5 = 440 + 870 = 1310$ ✓. (b) Nothing changes: B uses only 15 cameras, so with 17 cameras B is still feasible and still optimal, with 1300. A camera is worth 0 because some cameras are left over.

Q2 (Part 2). Prices $y_1, y_2, y_3 \geq 0$ for the resources R_1, R_2, R_3; read the columns of A:

$$\begin{aligned} \min\ w = 5y_1 + 11y_2 + 8y_3 \quad \text{s.t.} \quad & 2y_1 + 4y_2 + 3y_3 \geq 5 \quad \text{(model 1)} \\ & 3y_1 + y_2 + 4y_3 \geq 4 \quad \text{(model 2)} \\ & y_1 + 2y_2 + 2y_3 \geq 3 \quad \text{(model 3)}, \qquad y \geq 0. \end{aligned}$$

The primal has 3 variables and 3 constraints, so the dual has 3 variables and 3 constraints.

Q3 (Part 3). The plan $(12, 12)$ uses $36 \leq 40$ assembly hours, $60 \leq 60$ testing hours, $12 \leq 18$ cameras: feasible, and it earns $480 + 720 = 1200$. The rents $(20, 10, 0)$: drone A gets $20 + 30 = 50 \geq 40$ ✓, drone B gets $40 + 20 + 0 = 60 \geq 60$ ✓: acceptable, with total rent $800 + 600 = 1400$. By weak duality, $1200 \leq z^* \leq 1400$. A plan that earns 1450 is impossible: every plan earns at most 1400. (In fact $z^* = 1300$.)

Q4 (Part 4). The 10 minutes bring about $10 \times \frac{9}{2} = 45$ thousand DA per day, and cost 30: a good deal, with a net gain of 15. Check: $2x_1 + 4x_2 = 490$ and $3x_1 + 2x_2 = 360$. Multiply the second by 2 and subtract the first: $4x_1 = 230$, so $x_1 = 57.5$ and $x_2 = (360 - 172.5)/2 = 93.75$. The sensors: $93.75 \leq 100$ ✓, so this plan is feasible; $z = 12 \times 57.5 + 20 \times 93.75 = 690 + 1875 = 2565 = 2520 + 45$ ✓.

Q5 (Part 5). Slacks: R_1: $4 + 0 + 1 = 5$, so $s_1 = 0$; R_2: $8 + 0 + 2 = 10$, so $s_2 = 1 > 0$ and $y_2 = 0$; R_3: $6 + 0 + 2 = 8$, so $s_3 = 0$. Then $x_1 > 0$: $2y_1 + 4y_2 + 3y_3 = 5$, that is $2y_1 + 3y_3 = 5$; $x_3 > 0$: $y_1 + 2y_2 + 2y_3 = 3$, that is $y_1 + 2y_3 = 3$. From the second, $y_1 = 3 - 2y_3$; then $6 - 4y_3 + 3y_3 = 5$, so $y_3 = 1$ and $y_1 = 1$. Check $y = (1, 0, 1)$: $y \geq 0$ ✓, and the dual constraint of x_2: $3 + 0 + 4 = 7 \geq 4$ ✓. So $x = (2, 0, 1)$ is optimal, with $z^* = 10 + 3 = 13 = w^* = 5 + 8$. Shadow prices: 1 for R_1, 0 for R_2 (one unit unused), 1 for R_3.

Q6 (Part 6). A minimisation: read Rule 5.1 from right to left. Constraint 1 is "$\geq$" (natural in a min): $y_1 \geq 0$. Constraint 2 is "$\leq$" (strange in a min): $y_2 \leq 0$. The variables are ≥ 0: the dual constraints are "$\leq$".

$$\max\ w = 3y_1 + y_2 \quad \text{s.t.} \quad y_1 + y_2 \leq 4,\ \ y_1 - y_2 \leq 3,\ \ y_1 \geq 0,\ y_2 \leq 0.$$

At $(0, 3)$: constraint 2 gives $0 - 3 = -3 < 1$, not binding, so $y_2 = 0$. $x_2 = 3 > 0$: the dual constraint of x_2 is an equality, $y_1 - y_2 = 3$, so $y_1 = 3$. Check: $y_1 + y_2 = 3 \leq 4$ ✓, $y_1 \geq 0$, $y_2 \leq 0$ ✓. So $y = (3, 0)$ and $w^* = 9 = z^*$ ✓.